Words That Make Language Models Perceive

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Abstract

Large language models (LLMs) trained purely on text ostensibly lack any direct perceptual experience, yet their internal representations are implicitly shaped by multimodal regularities encoded in language. We test the hypothesis that explicit sensory prompting can surface this latent structure, bringing a text-only LLM into closer representational alignment with specialist vision and audio encoders. When a *sensory prompt* tells the model to 'see' or 'hear', it cues the model to resolve its next-token predictions as if they were conditioned on latent visual or auditory evidence that is never actually supplied. Our findings reveal that lightweight prompt engineering can reliably activate modality-appropriate representations in purely text-trained LLMs.
Project page: sophielwang.com/sensory
Code: github.com/sophicle/sensory

1 Introduction

On its face, predicting the next word in web text appears orthogonal to perception. Language contains descriptions, but not the sensations themselves. Patel & Pavlick [2022] highlighted the difficulty of directly encoding the meaning of sensory inputs using language alone. This tension echoes the *symbol-grounding problem*, which asks how purely textual symbols can acquire intrinsic meaning without being anchored in direct perceptual experience [Harnad, 1990].

For LLMs, this becomes a question of whether they are merely manipulating surface statistics of text or encoding knowledge that connects text to the sensory world (i.e., are LLMs grounded?). One way to test this is by measuring how closely their embeddings align with those of models trained explicitly on sensory data. By defining the meaning of a symbol through the relationships it maintains with others [Wittgenstein, 1953], alignment can be quantified through *kernel-based representational similarity metrics* (e.g., mutual k-nearest neighbors). In this view, if the geometry of an LLM's representations resembles that of a vision model, then it encodes text in a way that is closer to the visually grounded representation. Huh et al. [2024] demonstrated that as models become more capable in their respective modalities, their kernel structures become more similar. They argue that this convergence reflects the existence of a shared latent structure underlying different modalities.

While such cross-modal convergence emerges with scale, it raises an interesting question. Instead of treating alignment as a fixed property of a model, can we elicit it at inference time? And if so, can even text-only models be controllably steered into perceptually grounded representations? Our results suggest that the answer to both is yes. We find that:

> A simple cue like asking the model to 'see' or 'hear' can push a purely text-trained language model towards the representations of purely image-trained or purely-audio trained encoders.

A model's representation can be understood as the embeddings it assigns to a set of inputs. Typically, these are taken from single forward passes. In this work, we introduce the notion of *generative representations*: when an LLM is asked to generate, each output token involves another forward pass, which recursively builds a representation that is not only a function of the prompt, but also of the sequence generated so far. We observe that these autoregressive steps yield a representation that is more similar in geometry to an encoder that was trained on the corresponding modality. Moreover, we can control this generative representation; with an added sensory prompt, the resulting representation yields even higher alignment. To explain this intuitive, yet unexpected effect (Figure 1), we posit that an LLM implicitly maintains uncertainty over the kinds of evidence—visual, auditory, or otherwise—that could have produced the text it is reading. When the context begins with an explicit cue such as 'see' or 'hear', the model conditions its generations on a specific sensory

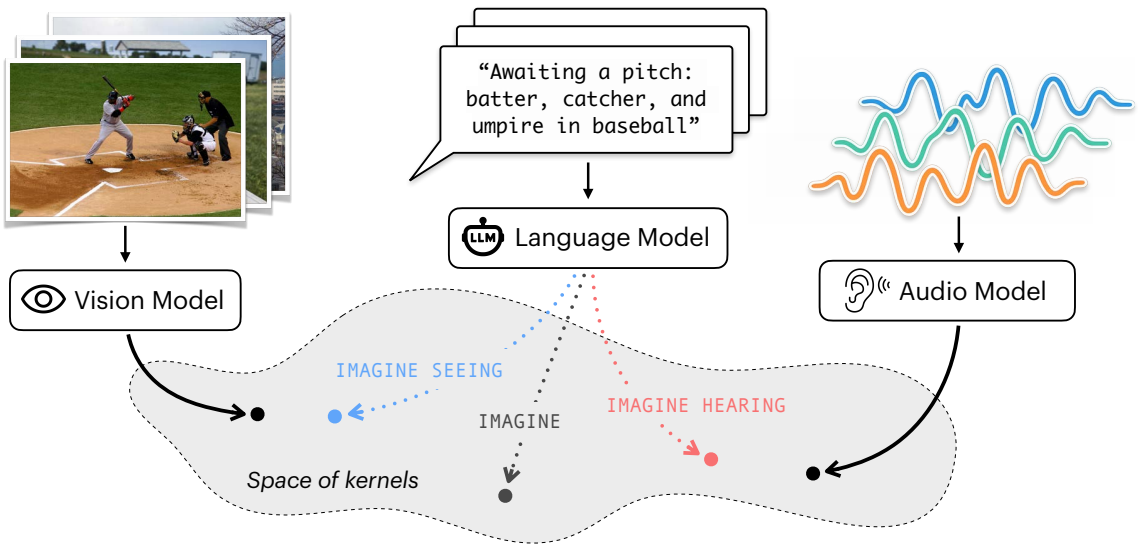

Figure 1: A cue that asks the model to 'see' (or 'hear') the provided text description moves the kernel representation of the model closer to the specialist model given the image (or audio) modality.

interpretation of the context. This means that representations are not fixed by training, but can be refined as a model reasons. In other words, LLMs can learn perceptually grounded representations from text alone, but we just need to know how to elicit it.

We quantify how sensory prompting steers the representation of an LLM by comparing them to frozen unimodal encoders in vision and audio domains. In our results, we find that:

- A single sensory word in the prompt can, through generation, shift the kernel of a text-only LLM closer to the geometry of sensory encoders.
- Representational similarity increases with generation length, as longer continuations give the model more opportunity to elaborate modality-specific content.
- Larger models exhibit higher alignment under sensory prompting and stronger modality separation.
- Visual cues allow LLMs to perform better on VQA in the text modality.

2 Methods

We evaluate how sensory prompts change the geometry of representations produced by text-only LLMs to resemble those of unimodal vision and audio encoders. To capture what the model represents as it generates, we incorporate generation into the representation. We then compare text- and sensory-induced kernels on paired datasets to quantify alignment, and we extend the analysis across additional models and datasets (Appendix C).

2.1 Extracting Generative Representations

Instead of taking LLM embeddings from a single forward pass, we average hidden states from autoregressive continuations conditioned on a fixed prompt. With prompt p and caption c, we form the prefix

$$x_{0:t} = [p \| c \| y_{1:t}], \qquad h_t^{(\ell)} = f_{\text{text}}^{(\ell)}(x_{0:t}), \qquad y_{t+1} \sim \text{Decode}\left(h_t^{(L)}\right),$$

where y_t is the t-th generated token, $h_t^{(\ell)}$ is the hidden state of the final token in $x_{0:t}$ at layer ℓ, and L is the total number of layers.

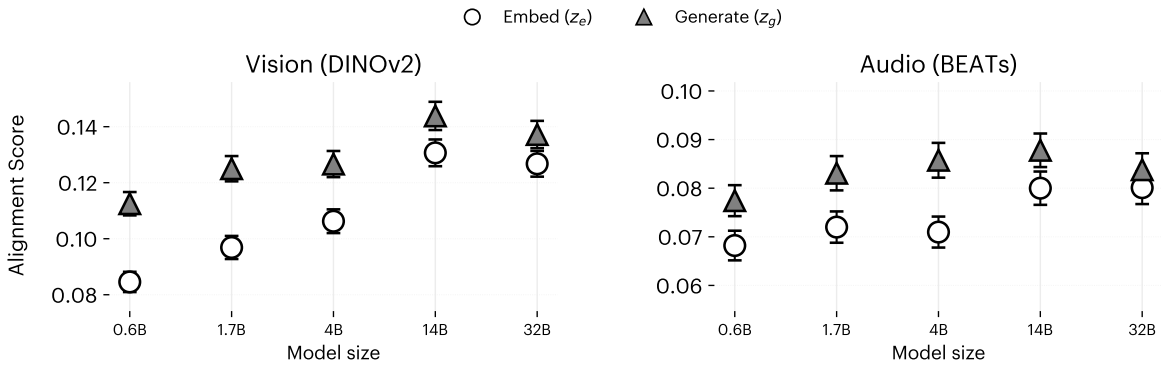

Figure 2: Generative representations (no sensory cue from Figure 3b, 128 token generations) yield higher alignment than single-pass embeddings.

We define the representations:

$$z_e^{(p)}(c) = \tfrac{1}{Lt}\sum_{\ell=1}^{L}\sum_{i=0}^{t} h_i^{(\ell)}, \qquad z_g^{(p)}(c,T) = \tfrac{1}{L(t+T)}\sum_{\ell=1}^{L}\sum_{i=t}^{t+T} h_i^{(\ell)}.$$

Here $z_e^{(p)}$ is the single-pass embedding from the initial caption, while $z_g^{(p)}$ averages across all tokens of the final continuation, over all layers. Like z_e, note that z_g maps a caption to a vector, but incorporates autoregressive hidden states as part of the representation.

Residual connections in the LLM architecture make this averaging a meaningful summary of the model's overall state, which we evaluate in Appendix A.

2.2 Quantifying Representational Similarity

Following the Platonic Representation Hypothesis framework [Huh et al., 2024], we define a *representation* as the set of embeddings a model produces on a dataset, and its induced *kernel* as the similarity structure among these embeddings. Given embeddings $\{z_i\}_{i=1}^n$, we define a kernel by $K_{ij} = \cos(z_i, z_j)$ and define $N_k^K(i)$ as the top-k neighbors of i under K. To compare two kernels K, K', we use mutual-kNN alignment,

$$\mathrm{Align}(K,K') = \frac{1}{n}\sum_{i=1}^{n}\frac{|N_k^K(i) \cap N_k^{K'}(i)|}{k},$$

where higher scores indicate higher representational similarity, i.e., two models are more aligned. For each prompt condition and dataset, we embed all samples, construct kernels from cosine neighbors, and compute alignment between the LLM and the corresponding sensory encoder. Error bars in paper figures denote ±1 bootstrap standard error ($B = 1000$), obtained by resampling N paired rows with replacement from the dataset to form bootstrap replicates and recomputing the mutual-kNN alignment score. This captures the variability of the score under resampling of the data.

2.3 Models

All models are kept frozen during evaluation.

Sensory Encoders: For vision, we use DINOv2-Base (ViT-B/14, 768-dim) [Oquab et al., 2023], a self-supervised model trained only on images. For audio, we use BEATs-Iter3 [Chen et al., 2022], a self-supervised model trained only on natural sounds (AudioSet).

Language Models: We evaluate frozen Qwen3 LLMs [Yang et al., 2025] across scales (0.6B, 1.7B, 4B, 8B, 14B, 32B). These models are trained only on text, with no vision or audio supervision.